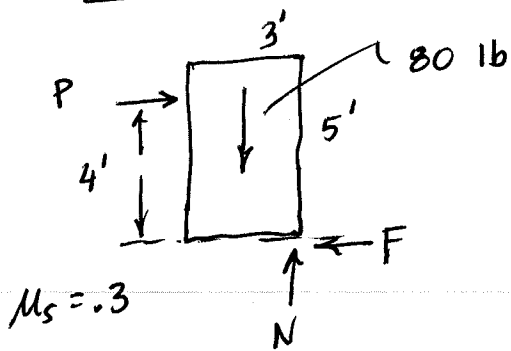


Friction : Example Problems

① Crate Slip/Tip



At what force P will this refrigerator move?

[By "move" the problem means slip or tip!!
Must check both!!]

Slip:

$$\uparrow N = 80 \text{ lb}$$

$$F = \mu N = .3(80) = 24 \text{ lb}$$

$$P_{\text{slip}} = F = 24 \text{ lb}$$

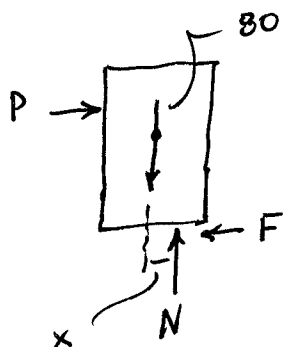
Tip: Draw N and F at corner where tip occurs. $\sum M_{\text{corner}} = 0$.

$$+\circlearrowleft \sum M_N = 0; \quad P(4) - 80(1.5) = 0$$

$$P_{\text{Tip}} = \frac{120}{4} = 30 \text{ lb}$$

Slip will occur first, at $P = 24 \text{ lb}$

Note about Hibbeler:



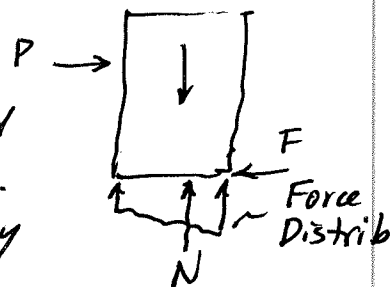
For this kind of problem, Hibbeler draws the N offset by x , as shown at left.

He's technically correct for the general slip case.

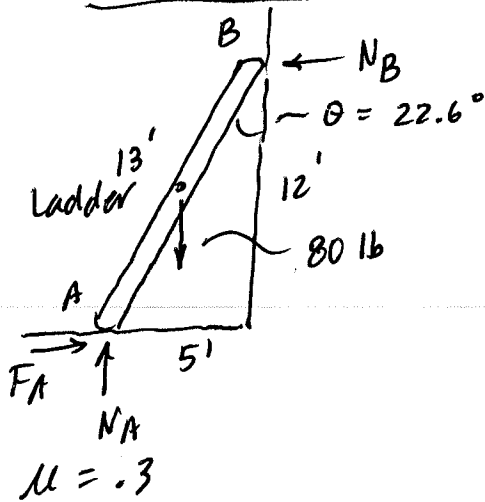
N is the resultant of a distributed load on bottom of crate.

N technically acts at corner ONLY for the tip case.

I do not use the x . I just solve both tip + slip cases. Easier!



② Ladder Problem



Given 60 lb ladder.

Smooth wall at B.

$\mu = .3$ on ground.

Will the ladder slip?

★ Do not set $F = \mu N$.

Can solve for F_A , N_A and N_B

from equil eqns.

Then check $\mu_s \leq \left(\frac{F_A}{N_A} \right)$?

$$+\circlearrowleft \sum M_A = 0; \quad 80(2.5) - N_B(12) = 0$$

$$N_B = 16.67 \text{ lb}$$

$+\uparrow$

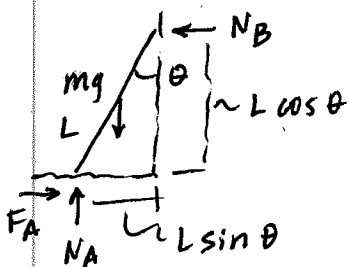
$$N_A = 80 \text{ lb}$$

$\rightarrow$

$$F_A = N_B = 16.67 \text{ lb}$$

Check: $\mu = .3 \leq \frac{F_A}{N_A} = \frac{16.67}{80} = .208$

Slip will not occur, ladder will remain in equilibrium.



At what θ will ladder slip (for a given μ)?

$$+\circlearrowleft \sum M_A = 0; \quad mg\left(\frac{L}{2} \sin \theta\right) - N_B(L \cos \theta) = 0$$

$$\frac{mg}{2} \sin \theta = N_B \cos \theta$$

$$\rightarrow F_A = \mu N_A = N_B$$

$$\frac{mg}{2} \sin \theta = \mu N_A \cos \theta$$

$$\uparrow N_A = mg$$

$$\tan \theta = 2\mu$$

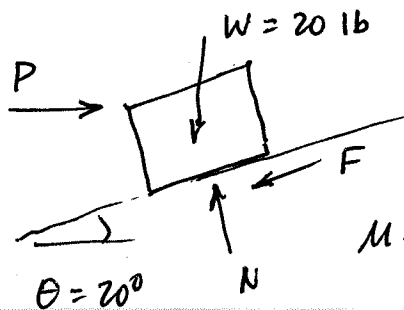
$$\theta = \tan^{-1} 2\mu$$

For $\mu = .3$

$$\theta = \tan^{-1} .6$$

$$\theta = 31^\circ$$

③ Crate on Slope



A 20 lb box rests on a slope, $\theta = 20^\circ$.

Determine P when:

(a) $F = 0$,

(b) When $P = 10$ lb, find F.

(c) When $P = 5$ lb, find F.

(d) Find P when motion impends upslope. ($\mu = .2$)

(e) Find P when motion impends downslope ($\mu = .2$)

$$+\uparrow N - W \cos 20 - P \sin 20 = 0$$

$$\boxed{N = W \cos 20 + P \sin 20}$$

$$+\rightarrow P \cos 20 - W \sin 20 - F = 0$$

$$\boxed{F = +P \cos 20 - W \sin 20}$$

See
Excel
file.

Given P, W, θ , can calc F and N.

(a) Excel Goal Seek: $\boxed{F = 0 \text{ when } P = 7.28 \text{ lb}}$

(b) When $P = 10$ lb, $\boxed{F = +2.56 \text{ lb}}$ Acts $\swarrow$

(c) When $P = 5$ lb, $\boxed{F = -2.14 \text{ lb}}$ Acts $\nearrow$

(d) $\boxed{P = 12.16 \text{ lb}}$ when $F = .2 N$ [Motion impends $\nearrow$]

(e) $\boxed{P = 3.06 \text{ lb}}$ when $F = .2 N$ [Motion impends $\swarrow$]